MANAN BHARDWAJ

Software Engineer | Backend Engineering | Machine Learning

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PROFESSIONAL SUMMARY

Full-stack software engineer focused on backend engineering and real-time, high-concurrency systems — using Node.js, TypeScript, WebSockets, and Redis to manage distributed state and prevent race conditions across concurrent clients. Also builds applied machine learning systems end to end, including a fine-tuned BERT scoring model and a from-scratch neural style transfer pipeline, across 5 self-built, production-style projects.

TECHNICAL SKILLS

- **Languages & Backend:** TypeScript, JavaScript, Python, Node.js, Express.js, Hono, REST API Design, Next.js, React
- **Real-Time & Concurrency:** WebSockets, WebRTC, Socket.IO, Concurrency Control, Race Condition Handling
- **Databases & Caching:** PostgreSQL, MongoDB, Redis (Atomic NX/EX Locking), Prisma ORM, Schema Design
- **Machine Learning / AI:** PyTorch, TensorFlow, Scikit-Learn, spaCy, BERT (Fine-Tuning), NLP, Computer Vision, CNN, LangChain, RAG, NumPy, Pandas
- **Cloud & DevOps:** AWS EC2, AWS S3, Docker, Kubernetes, Cloudflare, OAuth, JWT, Git
- **Core CS:** Data Structures and Algorithms

PROJECTS

NexusRoom | LIVE DEMO — TypeScript, WebSockets, WebRTC, Prisma, Monorepo

- Architected a 2D multiplayer virtual space with a real-time movement engine over WebSockets, synchronizing player position and state across all connected clients.
- Enabled proximity-based WebRTC peer-to-peer video calls, auto-triggered when two players enter detection range, layered on top of the distributed room-state engine with no added server-side media load.

SeatLock | LIVE DEMO — TypeScript, Express.js, Redis, Socket.IO, MongoDB, Cloudinary, JWT

- Engineered a high-concurrency seat reservation system, pushing live seat-lock state to all connected clients in real time via Socket.IO.
- Eliminated double-booking race conditions under concurrent access using atomic Redis NX/EX locking operations with a 5-minute lock expiry, instead of database-level locks.
- Implemented JWT-based authentication using short-lived access tokens and longer-lived refresh tokens, with a dedicated refresh endpoint issuing new access tokens automatically so users stayed logged in without repeated logins, gating seat booking behind required authentication.

ResumeIQ | LIVE DEMO — Python, spaCy, BERT (Fine-Tuned), FastAPI, NLP

- Developed an ATS-style resume analyzer that parses a resume end to end and generates a structured report covering strengths, skill gaps, and scoring feedback.
- Fine-tuned a pretrained BERT model on a custom-labeled resume dataset for contextual scoring, served via a FastAPI inference engine with spaCy handling entity extraction.

ClassRoll AI | LIVE DEMO — Python, Streamlit, SVM, Computer Vision, Voice Embeddings

- Delivered a dual-portal (teacher/student) AI attendance platform in Streamlit with face-based and optional voice-based student registration.
- Automated group-photo attendance marking using SVM classification, tested on group photos of up to 8 students with correct deduplication of repeat appearances.

StyleFusion | LIVE DEMO — Python, PyTorch, CNN, VGG Encoder, AdaIN

- Constructed a neural style transfer pipeline pairing a pretrained VGG encoder with a custom decoder network trained entirely from scratch at 256×256 resolution over 13 training epochs.
- Integrated an AdaIN (Adaptive Instance Normalization) layer as the architectural core to align content and style feature statistics at inference time.

WORK EXPERIENCE

Freelance Backend Developer — Contract, Attendance Management Application — **March 2025**

- Designed the backend data schema for storing user profile data, photo uploads, and subject/course codes.
- Implemented controller-level backend logic handling user registration workflows.

EDUCATION & CERTIFICATIONS

ABES Institute of Technology, Ghaziabad, India

B.Tech in Computer Science (Artificial Intelligence) — **2022 - 2026**

Certifications: Web Development, Data Structures and Algorithms, and Machine Learning - Apna College; Full-Stack Web Development Cohort - Harkirat Singh